

SOFTWARE ENGINEERING

Group 54 Unit Test Evidence Report

BUPT International School

Teaching Assistant Recruitment System

REPORT SCOPE

JUnit 5 backend and handler-level verification

RESULT

103 / 103 PASS

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1. Document Control

Item	Value
System	BUPT International School TA Recruitment System
Tested version	Test_Version_02
Test type	Backend unit and handler-level API testing
Framework	JUnit 5
Last verified date	2026-05-24
Latest verified result	103 tests successful, 0 tests failed

2. Executive Summary

This document records the automated test evidence for the implemented TA, Module Organiser (MO), and Administrator functions. The test suite verifies application workflow rules, persistence to JSON-backed storage, file upload/download handling, session-based authentication, role authorisation, notifications, workload calculations, AI matching integration, and administrative operations.

The current suite contains 18 JUnit test classes and 103 test methods. Every test passed in the latest verified execution.

Measurement	Result
Test classes executed	18
Test methods found	103
Test methods started	103
Test methods successful	103
Test methods failed	0
Containers failed	0

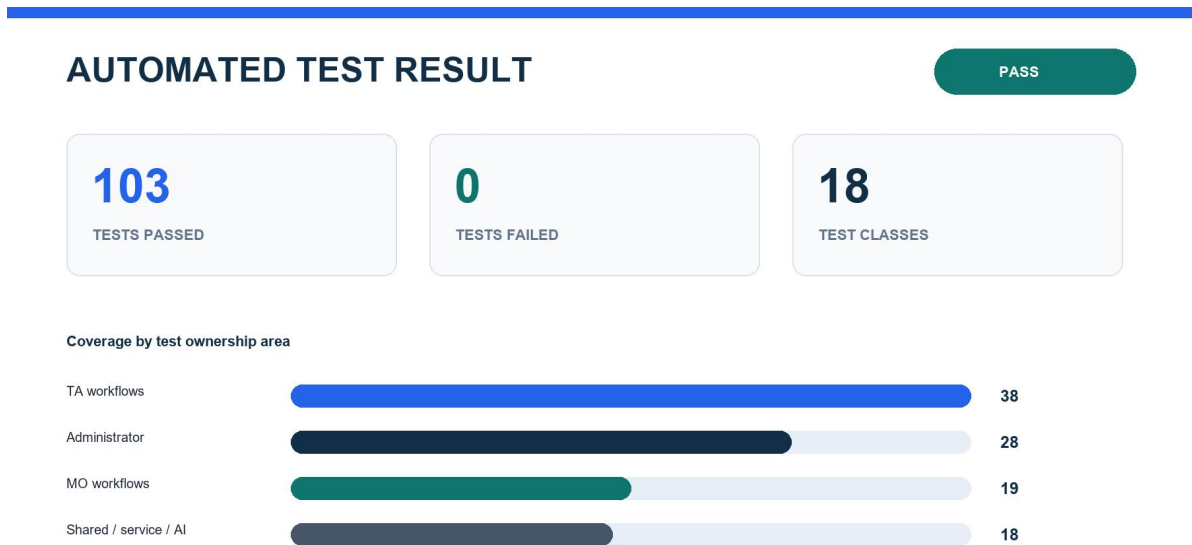


Figure 1. Verified test execution overview

3. Test Strategy

The application uses lightweight Java HTTP handlers and JSON file persistence rather than a relational database. The tests therefore operate at the handler and service boundary:

- [TestHttpExchange](#) simulates HTTP requests and captures status codes, response bodies, headers, and binary downloads.
- [DataService](#) is created with JUnit [@TempDir](#), giving each test isolated users, jobs, applications, drafts, notifications, uploads, settings, logs, and export task files.
- Authentication tests use actual session tokens created by the application's session service.
- Workflow tests invoke the same handlers registered by the running application, including [AuthHandler](#), [JobHandler](#), [ApplicationHandler](#), [DraftHandler](#), [UploadHandler](#), [NotificationHandler](#), and [AdminHandler](#).
- AI matching tests use the supported mock configuration so that results are deterministic and do not depend on an external network API.

HOW THE AUTOMATED TESTS VERIFY THE SYSTEM

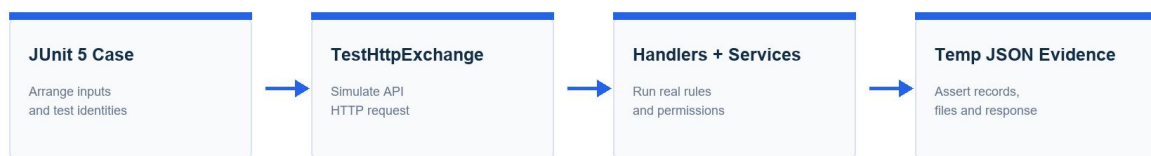


Figure 2. Handler-level unit testing method

4. Test Environment

Item	Evidence
Project folder	D:\Documents\GitHub\Software_Group_54_new\Test_Version_02
Source under test	src/main/java/com/bupt/tarecruit
Unit test source	src/main/java/com/bupt/tarecruit/unit_testing
Language/toolchain	Java / <code>javac</code> and <code>java</code>
Test framework jar	lib\junit-platform-console-standalone-1.10.2.jar
JSON dependency	lib\gson-2.10.1.jar
Persistent test data	Temporary JSON and upload folders provided by <code>@TempDir</code>
External database required	No
External mail server required	No
External AI service required	No for automated suite

5. Reproducible Execution

Run these commands in PowerShell. The final command prints each test with a pass mark in a tree, which is suitable for an evidence screenshot.

```
cd D:\Documents\GitHub\Software_Group_54_new\Test_Version_02

$main = "out\junit-main"
$test = "out\junit-test"
$root = (Resolve-Path .).Path
foreach ($p in @($main, $test)) {
    $full = Join-Path $root $p
    if (Test-Path -LiteralPath $full) {
        $resolved = (Resolve-Path -LiteralPath $full).Path
        if (-not $resolved.StartsWith($root)) {
            throw "Refusing to remove outside workspace: $resolved"
        }
        Remove-Item -LiteralPath $resolved -Recurse -Force
    }
    New-Item -ItemType Directory -Path $full -Force | Out-Null
}

javac -encoding UTF-8 -cp "lib\gson-2.10.1.jar" -d out\junit-main src\main\java\com\bupt\tarecruit\model\*.java
src\main\java\com\bupt\tarecruit\service\*.java src\main\java\com\bupt\tarecruit\handler\*.java
src\main\java\com\bupt\tarecruit\Main.java

javac -encoding UTF-8 -cp "lib\gson-2.10.1.jar;lib\junit-platform-console-standalone-1.10.2.jar;out\junit-main" -
d out\junit-test src\main\java\com\bupt\tarecruit\unit_testing\*.java

java -jar lib\junit-platform-console-standalone-1.10.2.jar execute --class-path "out\junit-test;out\junit-
main;lib\gson-2.10.1.jar" --scan-class-path --details=tree
```

If tree symbols do not render correctly in a Windows terminal, run `chcp 65001` before the final `java -jar` command.

6. User Story Coverage Matrix

6.1 Teaching Assistant Stories

Story ID	Story	Automated Evidence	Coverage Assessment
TA_01	Personal Profile Management	TaDataServiceStoriesTest	Profile update fields are persisted and returned.
TA_02	Vacancy Browsing and Filtering	MoJobManagementStoriesTest , TaJobBoundaryStoriesTest	Vacancy filters, posting visibility, and application counts are verified.
TA_03	Integrated Submission	TaDataServiceStoriesTest , AuthUploadDraftCoverageTest , TaJobBoundaryStoriesTest	PDF validation, byte storage/download, and uploaded CV filename linked to an application are verified.
TA_04	Self-Service Password Modification	TaAuthStoriesTest	Correct current password succeeds; incorrect password is rejected.
TA_05	Forgot Password Online Application	TaDataServiceStoriesTest , ExtendedAuthenticationAndAccessTest , AdSystemOperationsStoriesTest	Request storage, role-disambiguated processing, password reset, and notification are verified.
TA_06	Forgot Password Direct Admin Contact	AdminAdvancedOperationsStoriesTest	Administrative escalation path and password-reset notifications are verified at API level.
TA_07	Application Deadline Reminder	Notification infrastructure is covered by TaApplicationAccessAndNotificationTest and DataServiceIntegrityCoverageTest	Notification delivery mechanics are tested; timed 48-hour reminder scheduling and rendered banner remain integration/UI checks.

Story ID	Story	Automated Evidence	Coverage Assessment
TA_08	Document Preview Before Upload	AuthUploadDraftCoverageTest, TaDataServiceStoriesTest	Valid PDF upload/download data path is tested; browser preview rendering remains a UI check.
TA_09	Save as Draft	TaDataServiceStoriesTest, AuthUploadDraftCoverageTest, ExtendedAuthenticationAndAccessTest	Save, replace, retrieve, delete, dashboard listing, and TA-only access are verified.
TA_10	Application Status Tracking	TaApplicationAccessAndNotificationTest, MoApplicantReviewStoriesTest, DataServiceIntegrityCoverageTest	Role-scoped application access and status notification creation/read state are verified.
TA_11	Application Withdrawal	TaApplicationFlowTest	Own pending withdrawal succeeds; other-user and non-pending withdrawal are rejected.
TA_12	Preference Selection	TaApplicationFlowTest	Priority range, duplicate priority, maximum three active applications, reuse after inactive status, and reconciliation are verified.

6.2 Module Organiser Stories

Story ID	Story	Automated Evidence	Coverage Assessment
MO_01	Manage TA Work	MoJobManagementStoriesTest, MoApplicantReviewStoriesTest	Job creation/update/delete, applicant handling, cascaded deletion, and status decisions are verified.
MO_02	View Questionnaire Submissions	MoApplicantReviewStoriesTest	MO can view applicants and CV details only for owned postings.

Story ID	Story	Automated Evidence	Coverage Assessment
MO_03	User Registration and Role Selection	MoAuthStoriesTest , ExtendedAuthenticationAndAccessTest	MO account creation, portal login, and teacher ID persistence are verified.
MO_04	Simplify Approval Process	MoApplicantReviewStoriesTest	Approve/reject flows, invalid changes, quota protection, and TA notification are verified.
MO_05	Optimise UI Design	Not suitable for JUnit backend suite	Layout/navigation usability requires browser/manual or UI automation evidence.
MO_06	Applicant CV Review	MoApplicantReviewStoriesTest , TaJobBoundaryStoriesTest	Stored CV filename appears in applicant review and is linked from application submission.
MO_07	Applicant Selection and Approval	MoApplicantReviewStoriesTest	Approval/rejection and capacity limit enforcement are verified.
MO_08	Self-Service Password Modification	MoAuthStoriesTest	MO password change success and failure paths are verified.
MO_09	Forgot Password Online Application	MoAuthStoriesTest , ExtendedAuthenticationAndAccessTest , AdminBoundaryCoverageTest	MO request creation and correct role-specific administrative reset are verified.
MO_10	Forgot Password Direct Admin Contact	AdminAdvancedOperationsStoriesTest	Escalation and administrator notification behavior are verified.
MO_11	Support Different Course Types	MoJobManagementStoriesTest , DataServiceIntegrityCoverageTest	Different posting types and type-specific workload calculation are verified.

6.3 Administrator Stories

Story ID	Story	Automated Evidence	Coverage Assessment
AD_01	Monitor Overall TA Workload	AdWorkloadAndStatsStoriesTest , TaJobBoundaryStoriesTest , DataServiceIntegrityCoverageTest	Workload summary, overload detection, apply-time blocking, calculation and setting fallback are verified.
AD_02	Manage System Users	AdUserManagementStoriesTest , AdminBoundaryCoverageTest	Super-admin creation/update and standard-admin restrictions are verified.
AD_03	Oversee Job Postings	AdJobOversightStoriesTest	Admin update, delete, and application access for any posting are verified.
AD_04	View Basic Application Statistics	AdWorkloadAndStatsStoriesTest	Application/user totals, quota, priority, and open-job statistics are verified.
AD_05	Export Core System Data	AdSystemOperationsStoriesTest , AdminAdvancedOperationsStoriesTest	CSV creation, export-task history, download, and retry are verified.
AD_06	View System Key Event Logs	AdSystemOperationsStoriesTest , TaJobBoundaryStoriesTest	Audit-log search and workload-block audit generation are verified.
AD_07	Send Bulk Notifications	AdSystemOperationsStoriesTest , AdminBoundaryCoverageTest	Target-role delivery, inactive-user exclusion, and generated evidence file are verified.
AD_08	Configure Recruitment Cycle Dates	AdSystemOperationsStoriesTest	Setting update persistence is verified.
AD_09	Assign Basic User Roles	AdUserManagementStoriesTest , AdminAdvancedOperationsStoriesTest , AdminBoundaryCoverageTest	Role assignment, role-template CRUD/assignment count, and invalid template rejection are verified.

Story ID	Story	Automated Evidence	Coverage Assessment
AD_10	Admin Password Reset Processing	AdSystemOperationsStoriesTest , AdminAdvancedOperationsStoriesTest , AdminBoundaryCoverageTest , ExtendedAuthenticationAndAccessTest	Approve, reject, escalate, configurable initial password, role collision, and standard-admin restriction are verified.
AD_11	Search Users/Jobs	AdUserManagementStoriesTest , MoJobManagementStoriesTest	User and job search/filter behavior is verified.
AD_12	View Job Application Status Summary	AdWorkloadAndStatsStoriesTest	Priority and quota summary calculations are verified.
AD_13	View Individual User Details	AdUserManagementStoriesTest	User detail retrieval is verified.

7. Additional Implemented Feature Coverage

The implemented system contains behavior beyond the original user-story tables. These features are also represented in the automated suite.

Feature	Evidence
Authentication session invalidation on logout	AuthUploadDraftCoverageTest
Public prevention of administrator self-registration	AuthUploadDraftCoverageTest
Deactivated-account login blocking	ExtendedAuthenticationAndAccessTest
Portal login privacy for an identifier belonging to another role	TaAuthStoriesTest
Shared TA/MO ID disambiguation for login and password reset	TaAuthStoriesTest , ExtendedAuthenticationAndAccessTest
Notification ownership protection	ExtendedAuthenticationAndAccessTest
Upload/download authentication protection	ExtendedAuthenticationAndAccessTest
Job deletion cascade to applications, drafts, and notifications	MoJobManagementStoriesTest
Duplicate schedule-week validation	MoJobManagementStoriesTest

Feature	Evidence
Closed, expired, and full position application rejection	TaApplicationFlowTest , TaJobBoundaryStoriesTest
Weekly workload limit rejection with notification and audit log	TaJobBoundaryStoriesTest
Notification reconciliation idempotency	DataServiceIntegrityCoverageTest
Admin backup jobs and retry	AdminAdvancedOperationsStoriesTest
AI match explanation, saved result visibility, model validation, and usage limit	AiMatchingStoriesTest

8. Complete Test Class Inventory

The following inventory accounts for every test method in the successful 103-test execution.

Test Class	Tests	Primary Verified Area
TaDataServiceStoriesTest	9	Profile, password-reset request, drafts, upload validation and file persistence
TaAuthStoriesTest	7	TA login, role-separated portal access, password change, registration
TaApplicationFlowTest	11	Priority rules, deadlines, quota, reapplication, reconciliation, withdrawal
TaApplicationAccessAndNotificationTest	5	Application access boundaries and notification read operations
TaJobBoundaryStoriesTest	6	CV/application link, vacancy filters/status, workload rejection, posting/apply roles
MoAuthStoriesTest	5	MO registration, login, password change, password-reset request
MoJobManagementStoriesTest	8	Posting CRUD, schedule validation, deletion cascade, filtering, posting types
MoApplicantReviewStoriesTest	6	Applicant/CV access, decision actions, quota and invalid transitions
AdUserManagementStoriesTest	5	User creation, permissions, search, role/status update, details
AdJobOversightStoriesTest	3	Cross-posting administration and application visibility

Test Class	Tests	Primary Verified Area
AdWorkloadAndStatsStoriesTest	4	Workload and statistical dashboard calculations
AdSystemOperationsStoriesTest	5	CSV export, audit log search, broadcasts, settings, reset approval
AdminAdvancedOperationsStoriesTest	5	Role templates, backup/export tasks, reset rejection/escalation, super-admin scope
AdminBoundaryCoverageTest	6	Admin access restrictions, templates, reset template password, inactive broadcast exclusion
AuthUploadDraftCoverageTest	4	Admin registration restriction, logout, PDF download, draft endpoint
ExtendedAuthenticationAndAccessTest	6	Deactivation, teacher ID, shared-ID reset, ownership and authentication boundaries
DataServiceIntegrityCoverageTest	4	Notification reconciliation and workload calculation integrity
AiMatchingStoriesTest	4	AI matching outputs, persistence, supported model and usage quota
Total	103	All passed in latest execution

9. Complete Unit Test Design Catalogue

This section records the design of every JUnit test method currently executed by the suite. Each test is isolated with temporary data storage and is designed to verify one externally observable rule or persistence outcome.

9.1 TA Account, Profile, Upload And Draft Tests

Test Class	Test Method	Test Design	Expected Result
TaDataServiceStoriesTest	ta01_profileUpdateShouldPersistPersonalInformation	Submit changed TA personal fields through the profile flow and read stored user data.	Updated personal fields are persisted.
TaDataServiceStoriesTest	ta05_passwordResetRequestShouldBeStoredAsPending	Submit a TA password-reset request.	A reset record is stored with PENDING status.
TaDataServiceStoriesTest	ta09_saveDraftShouldOverwriteExistingDraftForSameUserAndJob	Save two drafts for the same TA and posting.	Only the latest draft content remains for that TA/posting pair.
TaDataServiceStoriesTest	ta09_deleteDraftShouldRespectNormalizedJobId	Save a draft and delete it using a job ID requiring normalisation.	The corresponding draft is removed.
TaDataServiceStoriesTest	ta09_dashboardShouldListOnlyCurrentUsersDraftsWithJobInfo	Store drafts for different TAs and request one TA's draft dashboard.	Only that TA's draft is returned, with posting details.
TaDataServiceStoriesTest	ta03_uploadStorageShouldSanitizeFileNameAndKeepBytes	Store an uploaded CV whose original name contains unsafe characters.	Stored filename is sanitised and downloaded bytes equal uploaded bytes.
TaDataServiceStoriesTest	ta03_uploadApiShouldRejectNonPdfFile	Attempt upload with a non-PDF extension.	Request is rejected and no invalid CV is accepted.
TaDataServiceStoriesTest	ta03_uploadApiShouldRejectInvalidBase64	Attempt PDF upload with invalid encoded content.	Request is rejected as an invalid upload.

Test Class	Test Method	Test Design	Expected Result
TaDataServiceStoriesTest	ta03_uploadApiShouldRejectEmptyFile	Attempt PDF upload without file content.	Request is rejected as an empty upload.
TaAuthStoriesTest	ta02_loginShouldAllowTaUsingStudentId	Log in through the TA portal using a valid student ID and password.	Login succeeds and a TA session token is returned.
TaAuthStoriesTest	ta02_loginShouldAllowTaWhenMOSHaresSameStudentId	Create TA and MO accounts sharing one ID, then log in via their respective portals.	Each portal authenticates the intended role only.
TaAuthStoriesTest	ta02_loginShouldNotRevealAccountExistsOnAnotherPortal	Submit valid TA credentials to the MO portal.	Login returns the generic MO invalid-credential response without role disclosure.
TaAuthStoriesTest	ta04_changePasswordShouldUpdateStoredPassword	Submit the correct current password and a new TA password.	Password update succeeds and the new value is stored.
TaAuthStoriesTest	ta04_changePasswordShouldRejectWrongOldPassword	Submit an incorrect current password.	Update is rejected and the previous password remains.
TaAuthStoriesTest	ta06_registerShouldCreateTaAccountAndReturnToken	Register a complete new TA account.	User is stored with role TA and an authenticated session is returned.
TaAuthStoriesTest	ta06_registerShouldRejectDuplicateUsername	Attempt TA registration with an existing username.	Request is rejected with conflict status and no duplicate is created.
AuthUploadDraftCoverageTest	uploadDownloadShouldReturnStoredPdfBytesForAuthenticatedUser	Authenticated user uploads and downloads a PDF.	Response content type is PDF and downloaded bytes match the upload.
AuthUploadDraftCoverageTest	draftHandlerShouldReturnEmptyDraftAndAllowUpsertThenGet	Read a missing draft, then save and retrieve it.	Empty response is explicit; saved letter and priority are returned.

9.2 TA Vacancy, Application And Notification Tests

Test Class	Test Method	Test Design	Expected Result
TaApplicationAccessAndNotificationTest	ta07_listApplicationsShouldReturnOnlyCurrentTasOwnApplications	Store applications belonging to multiple TAs and list as one TA.	Only the authenticated TA's applications are visible.
TaApplicationAccessAndNotificationTest	ta08_getApplicationDetailShouldAllowOwnApplication	Request detail of the authenticated TA's application.	Application detail is returned.
TaApplicationAccessAndNotificationTest	ta08_getApplicationDetailShouldRejectOtherTasApplication	Request another TA's application detail.	Access is denied.
TaApplicationAccessAndNotificationTest	ta10_notificationsShouldListOnlyUsersNotificationsAndMarkSingleAsRead	Store notifications for different users, list and read one as the owner.	Only own notifications are listed and selected item becomes read.
TaApplicationAccessAndNotificationTest	ta10_notificationsShouldMarkAllAsRead	Store multiple unread notifications for a TA and invoke read-all.	All of that TA's notifications become read.
TaApplicationFlowTest	ta12_applyShouldRejectPriorityOutsideRange	Submit application with preference outside 1..3.	Application is rejected.
TaApplicationFlowTest	ta12_applyShouldRejectDuplicatePriorityAmongActiveApplications	Apply using a priority already assigned to another active application.	Duplicate preference is rejected.
TaApplicationFlowTest	ta12_applyShouldRejectWhenThreeActiveApplicationsAlreadyExist	Submit a fourth active application.	Application is rejected because the active limit is three.
TaApplicationFlowTest	ta12_applyShouldRejectExpiredJobDeadline	Apply to a posting with a passed deadline.	Application is rejected and no record is added.
TaApplicationFlowTest	ta12_applyShouldRejectJobThatReachedQuota	Apply after an existing approved application fills the posting quota.	Application is rejected because the position is full.

Test Class	Test Method	Test Design	Expected Result
TaApplicationFlowTest	ta12_applyShouldAllowReusingPriorityFromWithdrawnOrRejectedApplications	Reuse priority values belonging only to inactive applications.	New pending application is accepted.
TaApplicationFlowTest	ta12_applyShouldAllowReapplyingSameJobAfterRejection	Apply again for a posting following a prior rejection.	New pending application is accepted.
TaApplicationFlowTest	ta12_reconcileShouldWithdrawDuplicateActivePriorityRecords	Persist conflicting active priorities then run reconciliation.	One conflicting record is withdrawn and active priorities are valid.
TaApplicationFlowTest	ta11_withdrawShouldAllowOwnApplication	TA withdraws their own pending application.	Status changes to WITHDRAWN .
TaApplicationFlowTest	ta11_withdrawShouldRejectNonPendingApplication	TA attempts withdrawal after an application is no longer pending.	Request is rejected and status is unchanged.
TaApplicationFlowTest	ta11_withdrawShouldRejectOtherUsersApplication	TA attempts to withdraw another TA's application.	Access is denied and status is unchanged.
TaJobBoundaryStoriesTest	applicationShouldLinkUploadedCVAndNotifyJobOwner	Upload a PDF and submit an application carrying the stored filename.	Application references the CV and MO receives new-application notification.
TaJobBoundaryStoriesTest	taShouldNotApplyForClosedPosition	Attempt application for a closed posting.	Request is rejected and no application is stored.
TaJobBoundaryStoriesTest	vacancyListShouldFilterTypeAndStatusAndExposeApplicationCounts	Query vacancies by type and status after applications exist.	Only matching postings are returned with correct application count.
TaJobBoundaryStoriesTest	applicationShouldBeBlockedWhenApprovedWorkloadWouldExceedWeeklyLimit	Give TA approved workload at the limit, then apply for further hours.	Application is blocked, and workload notification and audit log are stored.

Test Class	Test Method	Test Design	Expected Result
TaJobBoundaryStoriesTest	taShouldNotCreateVacancyPosting	Submit job-creation request authenticated as TA.	Posting is rejected because TA lacks posting permission.
TaJobBoundaryStoriesTest	moShouldNotSubmitTaApplication	Submit application request authenticated as MO.	Application is rejected because only TAs may apply.

9.3 MO Account, Posting And Selection Tests

Test Class	Test Method	Test Design	Expected Result
MoAuthStoriesTest	mo03_registerShouldCreateMoAccount	Register a new module organiser account.	MO user is stored successfully.
MoAuthStoriesTest	mo03_loginShouldAllowMoPortalAccess	Log in to the MO portal with valid MO credentials.	Login succeeds and MO access is returned.
MoAuthStoriesTest	mo08_changePasswordShouldUpdateStoredPassword	Authenticated MO submits correct current password and a new password.	Stored password changes.
MoAuthStoriesTest	mo08_changePasswordShouldRejectWrongOldPassword	Authenticated MO submits an incorrect current password.	Request is rejected and stored password remains unchanged.
MoAuthStoriesTest	mo09_forgotPasswordShouldCreateResetRequest	MO submits forgot-password information.	Pending password-reset request is stored.
MoJobManagementStoriesTest	mo01_createJobShouldAllowMoToPostTask	Authenticated MO creates a vacancy with requirements.	Open posting is stored under the MO owner.
MoJobManagementStoriesTest	mo01_updateJobShouldAllowOwnerToEditPostedTask	Owning MO edits posting fields and status.	Updated values are persisted.
MoJobManagementStoriesTest	mo01_createJobShouldRejectDuplicateCourseScheduleWeeks	Create course posting containing duplicate week entries.	Posting creation is rejected.

Test Class	Test Method	Test Design	Expected Result
MoJobManagementStoriesTest	mo01_updateJobShouldRejectDuplicateCourseScheduleWeeks	Edit a posting to contain duplicate week entries.	Posting update is rejected.
MoJobManagementStoriesTest	mo01_deleteJobShouldRejectNonOwner	A different MO attempts to delete an owned posting.	Access is denied and posting remains.
MoJobManagementStoriesTest	mo01_deleteJobShouldCascadeApplicationsDraftsAndApplicationNotifications	Delete a posting with related application, draft and notification records.	Posting and related records are removed.
MoJobManagementStoriesTest	mo01_listJobsShouldSupportPostedByAndSearchFiltering	Query postings by owner and search text.	Only matching posting is returned.
MoJobManagementStoriesTest	mo11_createJobShouldSupportDifferentCourseTypes	Create a non-course/alternative vacancy type.	Posting stores the selected type and course context.
MoApplicantReviewStoriesTest	mo06_viewApplicantsShouldReturnCvAndApplicantInfoForOwnJob	Owning MO lists applications for a posting with stored CV.	Applicant identity and CV filename are returned.
MoApplicantReviewStoriesTest	mo06_viewApplicantsShouldRejectOtherMosJob	MO requests applicants for another MO's posting.	Access is denied.
MoApplicantReviewStoriesTest	mo07_approveApplicantShouldUpdateStatusAndNotifyTa	Owning MO approves a pending applicant.	Application becomes APPROVED and TA receives notification.
MoApplicantReviewStoriesTest	mo07_approveApplicantShouldRejectWhenQuotaIsFull	MO attempts approval when another approved applicant fills quota.	Request is rejected and pending status remains.
MoApplicantReviewStoriesTest	mo07_rejectApplicantShouldUpdateStatusAndNotifyTa	Owning MO rejects a pending applicant.	Application becomes REJECTED and TA receives notification.
MoApplicantReviewStoriesTest	mo07_shouldRejectInvalidMoStatusChange	MO tries an unsupported status transition.	Request is rejected and pending status remains.

9.4 Administrator Tests

Test Class	Test Method	Test Design	Expected Result
AdUserManagementStoriesTest	ad02_superAdminShouldCreateUser	Super administrator creates a managed user.	New user is persisted.
AdUserManagementStoriesTest	ad02_nonSuperAdminShouldNotCreateUser	Standard administrator attempts user creation.	Request is forbidden.
AdUserManagementStoriesTest	ad09_superAdminShouldUpdateRoleAndActiveStatus	Super administrator updates user role and activation state.	Role/status changes are persisted.
AdUserManagementStoriesTest	ad11_userListShouldSupportSearchAndRoleFilter	Search and filter admin user list.	Result contains only matching users.
AdUserManagementStoriesTest	ad13_shouldViewIndividualUserDetails	Administrator requests a specific user's detail.	User detail is returned.
AdJobOversightStoriesTest	ad03_adminShouldUpdateAnyJobPosting	Administrator edits a posting owned by an MO.	Posting update is persisted.
AdJobOversightStoriesTest	ad03_adminShouldDeleteAnyJobPosting	Administrator deletes an MO posting.	Posting is removed.
AdJobOversightStoriesTest	ad03_adminShouldViewApplicationsForAnyJob	Administrator requests applications for an MO posting.	Applications are visible to admin.
AdWorkloadAndStatsStoriesTest	ad01_workloadShouldShowOverloadedTaAndSupportStatusFilter	Build overloaded approved workload and query admin workload with filter.	Overloaded TA appears in matching result.
AdWorkloadAndStatsStoriesTest	ad04_statsShouldReturnApplicationAndUserSummary	Request statistics after storing users and applications.	Summary totals are returned.
AdWorkloadAndStatsStoriesTest	ad12_statsShouldIncludePriorityAndQuotaSummary	Request statistics for prioritised applications and posting quota.	Priority and capacity summary values are returned.

Test Class	Test Method	Test Design	Expected Result
AdWorkloadAndStatsStoriesTest	ad04_statsOpenJobsShouldReflectAllLiveJobsNotOnlyCreatedInPeriod	Query period statistics while an older job remains open.	Live open-job count includes all currently open postings.
AdSystemOperationsStoriesTest	ad05_exportShouldCreateExportTaskAndCsvFile	Administrator requests core-data export.	Completed export task and CSV file are created.
AdSystemOperationsStoriesTest	ad06_auditLogsShouldSupportSearch	Store audit entries and search by detail text.	Only matching audit entry is returned.
AdSystemOperationsStoriesTest	ad07_bulkNotificationShouldSendToTargetRoles	Super administrator sends notification to selected roles.	Target TA/MO recipients receive the announcement.
AdSystemOperationsStoriesTest	ad08_superAdminShouldUpdateSystemSettings	Super administrator updates workload and cycle-date settings.	Settings are persisted.
AdSystemOperationsStoriesTest	ad10_approvePasswordResetShouldResetPasswordAndNotifyUser	Super administrator approves a pending reset request.	Initial password is assigned and user notification is created.
AdminAdvancedOperationsStoriesTest	adminRoleTemplatesShouldSupportCreateListDetailUpdateDeleteAndAssignedCount	Execute complete role-template lifecycle with assigned admin.	CRUD works and detail reports assignment count.
AdminAdvancedOperationsStoriesTest	backupTasksShouldCreateListDownloadAndRetryBackupZip	Create backup, list/download it, mark failed and retry.	ZIP is created/downloaded and retry completes.
AdminAdvancedOperationsStoriesTest	exportTasksShouldListDownloadAndRetryFailedExports	Store failed export, filter it, retry, and download output.	Retry produces downloadable completed CSV task.
AdminAdvancedOperationsStoriesTest	passwordResetRejectAndEscalateShouldPersistReasonAndNotifySuperAdmin	Reject reset with reason and escalate a pending request.	Reason is stored and escalation notification is sent.
AdminAdvancedOperationsStoriesTest	standardAdminShouldBeBlockedFromSuperAdminOperations	Standard admin attempts settings change and bulk broadcast.	Both super-admin-only operations are forbidden.

Test Class	Test Method	Test Design	Expected Result
AdminBoundaryCoverageTest	nonAdminShouldBeDeniedAdminStatistics	TA calls administrator statistics endpoint.	Access is forbidden.
AdminBoundaryCoverageTest	standardAdminShouldReadUsersButNotDeleteAccounts	Standard admin reads user list then attempts deletion.	Read succeeds; destructive write is forbidden.
AdminBoundaryCoverageTest	superAdminShouldRejectUnknownAdminRoleTemplateAssignment	Create admin user referencing non-existent role template.	Request is rejected and user is not created.
AdminBoundaryCoverageTest	activePasswordResetTemplateShouldControlApprovedInitialPassword	Configure active email template with custom reset password, then approve request.	User receives configured initial password.
AdminBoundaryCoverageTest	bulkNotificationShouldSkipInactiveRecipientsAndCreateEvidenceFile	Broadcast to TA role containing active and inactive accounts.	Active user receives notice, inactive user is skipped, evidence file exists.
AdminBoundaryCoverageTest	standardAdminShouldNotProcessPasswordResetRequest	Standard administrator attempts approval of TA reset.	Request remains pending and password unchanged.

9.5 Shared Access, Service Integrity And AI Tests

Test Class	Test Method	Test Design	Expected Result
AuthUploadDraftCoverageTest	authShouldRejectSelfRegisteredAdminAccounts	Submit public registration request with role ADMIN .	Registration is forbidden and no admin account is created.
AuthUploadDraftCoverageTest	authLogoutShouldInvalidateSessionToken	Authenticate, logout, then reuse old token for current-user request.	Token no longer authorises access.
ExtendedAuthenticationAndAccessTest	loginShouldRejectDeactivatedAccountWithoutIssuingAccess	Attempt valid login for deactivated TA.	Login is forbidden and token is not issued.

Test Class	Test Method	Test Design	Expected Result
ExtendedAuthenticationAndAccessTest	moRegistrationShouldPersistTeacherIdAndExposeItFromCurrentUser	Register MO with teacher ID.	ID is stored and exposed as teacherId in authenticated response.
ExtendedAuthenticationAndAccessTest	passwordResetRoleShouldTargetMOWhenTaAndMoShareIdentifier	TA and MO share an ID; process reset specifically for MO role.	MO password resets while TA password remains unchanged.
ExtendedAuthenticationAndAccessTest	notificationShouldRejectReadingAnotherUsersItem	Different TA attempts to mark owner's notification read.	Request is forbidden and read flag remains false.
ExtendedAuthenticationAndAccessTest	draftEndpointsShouldBeRestrictedToTaUsers	MO attempts to save an application draft.	Request is forbidden and no draft is saved.
ExtendedAuthenticationAndAccessTest	uploadAndDownloadShouldRequireAuthentication	Upload and download file without session token.	Both operations reject unauthenticated request.
DataServiceIntegrityCoverageTest	notificationReconciliationShouldBackfillApplicationEventsOnlyOnce	Reconcile notifications twice after approved application data exists.	MO and TA workflow notifications each appear once only.
DataServiceIntegrityCoverageTest	structuredScheduleShouldCalculateHoursPerWeekFromSelectedPeriods	Supply structured selected periods across two weeks.	Weekly hours equal period count multiplied by configured duration.
DataServiceIntegrityCoverageTest	finalExamWorkloadShouldUseExamDurationAsSingleWorkloadBucket	Supply final-exam job with exam duration.	Duration is returned in single workload bucket.
DataServiceIntegrityCoverageTest	invalidWorkloadSettingShouldFallBackToDefaultLimit	Persist a non-numeric maximum-hour setting.	Service returns default limit of 20 hours.
AiMatchingStoriesTest	aiMatchShouldExplainMatchedAndMissingSkillsForTa	Request AI match for TA/job with mock matching enabled.	Response contains match explanation, skills and recommendation information.

Test Class	Test Method	Test Design	Expected Result
AiMatchingStoriesTest	moApplicationListShouldOnlyIncludeSavedAiMatchAfterRun	List MO applications before and after a stored AI matching run.	AI summary appears only after saved matching result exists.
AiMatchingStoriesTest	taAiMatchShouldBeLimitedToThreeSuccessfulCalls	TA invokes successful AI matching repeatedly.	First three succeed; further use is restricted.
AiMatchingStoriesTest	aiMatchShouldRejectUnsupportedModel	Request AI matching with an unsupported model identifier.	Request is rejected.

10. Selected Critical Test Design Details

The following cases were selected because they protect high-value workflows: identity separation, submission integrity, workload safety, selection fairness, privileged administration, and controlled AI use.

10.1 Shared Identifier Password Reset Targets The Correct Role

Item	Detail
Test method	<code>ExtendedAuthenticationAndAccessTest.passwordResetRoleShouldTargetMowhenTaAndMoShareIdentifier</code>
Risk addressed	A TA and an MO may use the same numeric identifier; a reset must not change the wrong person's password.
Preconditions	One TA and one MO user are stored with identifier SHARED-110, each with a distinct original password.
Input	Password-reset request with <code>role="MO"</code> and <code>teacherId="SHARED-110"</code> , followed by super-admin approval.
Execution	Call <code>/api/auth/password-reset</code> , locate stored MO reset request, then call <code>/api/admin/password-resets/{id}</code> with action <code>APPROVE</code> .
Assertions	Request is created and approved; TA password remains <code>ta-old</code> ; MO password becomes the initial reset password <code>123456</code> .
Importance	Demonstrates that role identification is enforced at a sensitive account-recovery boundary.

10.2 Uploaded CV Is Linked To The Submitted Application

Item	Detail
Test method	<code>TaJobBoundaryStoriesTest.applicationShouldLinkUploadedCvAndNotifyJobOwner</code>
Risk addressed	A resume could be uploaded but not associated with the application reviewed by the MO.
Preconditions	Authenticated TA, owning MO and an open vacancy exist.
Input	Base64 PDF upload followed by application JSON containing the returned <code>cvFileName</code> .
Execution	POST upload through <code>UploadHandler</code> ; POST application through <code>JobHandler</code> .

Item	Detail
Assertions	Upload succeeds; application record contains exactly the saved CV filename; MO receives a New Application notification naming the posting.
Importance	Verifies the end-to-end data relationship between uploads/ and applications.json used in applicant review.

10.3 Application Is Blocked When Weekly Workload Would Exceed Limit

Item	Detail
Test method	<code>TaJobBoundaryStoriesTest.applicationShouldBeBlockedWhenApprovedWorkloadWouldExceedWeeklyLimit</code>
Risk addressed	A TA may be approved or apply for duties above the configured weekly workload threshold.
Preconditions	maxWeeklyHours is set to 2 ; TA already has an approved two-hour posting in week 1; another one-hour posting is open.
Input	Application for the additional posting with an unused priority.
Execution	POST <code>/api/jobs/{id}/apply</code> as the TA.
Assertions	Response is conflict status 409 ; body identifies week 1 overload; a WORKLOAD notification and WORKLOAD_BLOCK audit record are saved.
Importance	Protects workload policy and leaves visible evidence explaining why an application was refused.

10.4 MO Cannot Approve Beyond Posting Quota

Item	Detail
Test method	<code>MoApplicantReviewStoriesTest.mo07_approveApplicantShouldRejectWhenQuotaIsFull</code>
Risk addressed	An MO could approve more applicants than a posting permits.
Preconditions	Vacancy quota is filled by an existing approved TA; a second application remains pending.
Input	MO status-update request attempting to approve the pending application.
Execution	PUT <code>/api/applications/{id}/status</code> with APPROVED .
Assertions	Response is 409 ; message indicates the quota is reached; pending application stays PENDING .

Item	Detail
Importance	Preserves recruitment capacity rules at the decision-making step.

10.5 Only Super Admin Can Process Password Reset Requests

Item	Detail
Test method	<code>AdminBoundaryCoverageTest.standardAdminShouldNotProcessPasswordResetRequest</code>
Risk addressed	A standard admin could otherwise gain account-control privileges reserved for the system administrator.
Preconditions	Standard administrator, TA account and pending reset request exist.
Input	Approval request authenticated with standard-admin session.
Execution	PUT <code>/api/admin/password-resets/{id}</code> with action <code>APPROVE</code> .
Assertions	Response is forbidden; request remains <code>PENDING</code> ; TA password remains unchanged.
Importance	Validates the difference between normal administrator visibility and super-administrator authority.

10.6 TA AI Matching Use Is Limited

Item	Detail
Test method	<code>AiMatchingStoriesTest.taAiMatchShouldBeLimitedToThreeSuccessfulCalls</code>
Risk addressed	Unrestricted AI matching calls may exceed intended usage limits or service cost controls.
Preconditions	TA and eligible vacancy exist; deterministic AI matching mode is enabled for testing.
Input	Repeated supported AI matching requests from the same TA.
Execution	Call the job matching endpoint four times.
Assertions	Three successful results are accepted; the fourth request is rejected by the per-TA limit.
Importance	Verifies additional implemented functionality not stated in the original user-story sheet.

11. Latest Successful Execution Evidence

The complete suite was compiled and executed on 2026-05-24 using the JUnit Platform Console standalone runner with `--details=tree`.

```
Test run finished after 3143 ms
[    21 containers found    ]
[     0 containers skipped  ]
[    21 containers started  ]
[     0 containers aborted  ]
[    21 containers successful ]
[     0 containers failed   ]
[   103 tests found        ]
[     0 tests skipped      ]
[   103 tests started      ]
[     0 tests aborted      ]
[   103 tests successful    ]
[     0 tests failed       ]
```

The tree-mode console output displays a pass mark beside each individual test method. A terminal screenshot after executing the command in Section 5 can be used as visual submission evidence.

12. Scope Limitations And Follow-Up Verification

The JUnit suite proves backend and handler behavior in an isolated local test environment. It does not replace the following checks:

Area	Recommended Verification
Page layout, responsive design, and user navigation	Manual browser review or UI automation
PDF preview rendering before submission	Browser-level test using an actual PDF
Deadline reminder scheduling and visible banner timing	Integration test with controlled system time
Real SMTP mail delivery	Test mail account or mocked mail transport integration test
Real external AI provider response and availability	Controlled staging test with secret supplied through configuration
Browser storage behavior, if used by frontend pages	Browser automation

13. Security Observations Identified During Test Review

These findings are outside the JUnit coverage result but should be addressed before deployment or demonstration with real credentials:

- SMTP connection credentials are stored in plaintext in [data/settings.json](#).
- An AI service API key is hardcoded in

[src/main/java/com/bupt/tarecruit/service/AiMatchingService.java](#).

- User password values are represented as plain strings in the JSON-backed model and should be replaced by securely hashed password storage for a production deployment.

Recommended action: revoke and rotate exposed credentials, load secrets from environment-specific configuration that is excluded from version control, and introduce password hashing with migration for existing accounts.